

Polarization by Aligned Grains: What SEDust and MoCafe Would Need

Abstract

This memo audits what it would take to replace the empirical dichroic polarization treatment of Seon (2018, ApJ 862, 87) with genuine aligned-grain physics in the current SEDust+MoCafe stack. The 2018 model is dissected first: its *scattering* polarization was already exact (full numerical Mie Mueller matrix with Stokes tracing), but its *dichroic* polarization was an add-on fitting formula with no alignment physics, no wavelength dependence, and additive rather than multiplicative accumulation of Stokes Q and U . The three ingredients that would be needed instead are then examined in turn: polarized cross sections, an alignment prescription, and a polarized transfer solver. The central practical finding is that the first of these is already in hand. The file `q_DH21Ad_P0.20_Fe0.00_1.400.dat.gz` stores three separate orientation blocks rather than an orientation average, so C_{pol} can be built today without any new T-matrix computation; values derived from it reproduce the Hensley & Draine (2023) released `polarized_extinction.dat` to four significant figures. A staged plan follows, in which stage 1 (SEDust cross sections and an emission accumulator) is a few days of low-risk work, stage 2 already permits a quantitative comparison against the SOFIA/HAWC+ 154 μm maps of NGC 891, and only stage 4 (self-consistent alignment) requires the invasive change of a vectorial radiation-field pass. Validation targets and the points that remain unverified are listed explicitly.

Contents

1. Motivation: anatomy of the 2018 model	3
1.1 What was already correct: scattering polarization	3
1.2 What was empirical: dichroic polarization	3
1.3 What was absent: alignment physics	4
1.4 The implementation that survives in the tree	4
2. What has changed since 2018	5
3. Polarized optical properties	6
3.1 The minimum required set	6
3.2 Modified picket fence approximation	6
3.3 Partial alignment and the Rayleigh reduction factor	6
3.4 Sign convention	7
3.5 The asset already in the tree	7
3.6 Wavelength coverage of the polarized optics	7
3.7 Dichroism below the Lyman limit	7
3.8 Index convention traps	8
3.9 HD23 alignment assumptions	8

4. What alignment theory hands to radiative transfer	9
4.1 The interface is one number	9
4.2 The analytic alignment radius	9
4.3 Rotational disruption	9
4.4 Magnetically enhanced alignment and its 2025 revision	10
4.5 Angle convention	10
4.6 Which axis do grains align with?	10
4.7 The polarization hole	11
5. Polarized transfer and its implementation	11
5.1 The Stokes transfer equation	11
5.2 How POLARIS assembles the matrix	11
5.3 Prefer the closed-form solution to adaptive integration	12
5.4 The three-pass structure and what it buys	12
5.5 The radiation field must be vectorial	12
5.6 Torque efficiency tables are not needed	13
5.7 Stochastic heating and polarization	13
6. Current state of the codes	13
6.1 SEDust	13
6.2 MoCafe v2.00	14
6.3 Reference sources in the tree	15
7. Staged proposal	15
8. What the update actually touches	16
8.1 SEDust, stage 1 (implemented)	16
8.2 MoCafe, stage 2	18
8.3 MoCafe, stage 3	19
8.4 MoCafe, stage 4	19
8.5 Order of work and what each stage buys	19
9. What optical polarized RT additionally requires	19
9.1 The stage table presupposes data that does not exist yet	20
9.2 Recommended order	23
9.3 What the Hoang group assumes	23
9.4 A working prescription	25
9.5 A constraint that applies only to a multi-wavelength extension	26

10. The aligned scattering matrix, as built	26
10.1 Why the phase matrix carries three angles	26
10.2 The meridional basis	27
10.3 Exact symmetries	27
10.4 Aligned plus unaligned, exactly	27
10.5 The extinction matrix from forward amplitudes	28
10.6 Where each quantity enters the Peest transfer	28
11. Validation targets	28
12. Unverified items	30
13. References	31

1. Motivation: anatomy of the 2018 model

The question this memo answers is narrow: if Seon (2018, ApJ 862, 87; 2018ApJ...862...87S), “Polarization as a Probe of Thick Dust Disks in Edge-on Galaxies: Application to NGC 891”, were redone now, what would have to be built? Answering it requires an exact statement of what that model did and did not contain. The description below is taken from arXiv:1806.06101v1.

1.1 What was already correct: scattering polarization

The scattering half of the 2018 calculation was not an approximation. Spherical grains were treated with the full numerical Mie Mueller matrix, retaining all four independent elements S_{11} , S_{12} , S_{33} , S_{34} ; photons carried a Stokes vector; the scattering and peel-off bookkeeping followed the algorithm of Peest et al. (2017, A&A 601, A92; 2017A&A...601A...92P); and the implementation was verified against the polarized scattering benchmark of Bianchi, Ferrara & Giovanardi (1996, ApJ 465, 127; 1996ApJ...465...127B). No Henyey–Greenstein or Rayleigh proxy was used for the phase matrix. That part of the code needs no revision.

1.2 What was empirical: dichroic polarization

The dichroic half descends from the prescriptions of Wood (1997, ApJ 477, L25; 1997ApJ...477L...25W) and Wood & Jones (1997, AJ 114, 1405; 1997AJ...114.1405W). Seon (2018) adopted the Whittet et al. (2008, ApJ 674, 304; 2008ApJ...674...304W) empirical relation $P_K/\tau_K \approx 5.08 \tau_V^{-0.52}$ and converted it to

$$P_V = 2.7 \tau_V^{0.48} \quad [\%], \quad (1)$$

then, at every interaction site, added a polarized fraction $P_V \sin^2 \theta$ oriented parallel to the local magnetic field projected on the sky, with θ the angle between the field and the photon direction. The author’s own summary in that paper is unambiguous: “we assumed spherical dust grains when calculating the Mueller matrix and adopted an empirical approach to take into account the dichroic polarization effect.”

The internal tension is worth stating plainly. Under a strictly spherical grain assumption the dichroic cross-section difference vanishes identically, so the dichroic term was necessarily grafted on from outside the optical model rather than derived from it.

1.3 What was absent: alignment physics

Radiative torque (RAT) alignment appears in the 2018 paper only as introductory background. Neither the Rayleigh reduction factor R nor an alignment size threshold a_{align} enters the calculation anywhere. All alignment efficiency is absorbed into the single exponent 0.48 and the normalization of Eq. (1), which are Milky Way averages. That is a double-counting hazard as well as a physics gap: the Whittet relation already folds in turbulent depolarization along real Galactic sight lines, plus a mean alignment efficiency and a mean grain composition, and the model then imposes its own explicit field geometry on top of that same quantity.

The magnetic field was treated as two separate deterministic runs, purely toroidal and purely vertical, compared against each other. There was no turbulent component and no field-strength parameter. Crucially, the field in that model sets only the polarization position angle and the $\sin^2 \theta$ projection factor; it has no influence on alignment efficiency, because no alignment efficiency is computed.

The dust model was MRN with Draine & Lee (1984, ApJ 285, 89; 1984ApJ...285...89D) and Draine (2003, ApJ 598, 1017; 2003ApJ...598.1017D) silicate and graphite optical constants, with PAHs excluded. The choice matters more than it looks: the maximum single-scattering polarization is 36% for that mixture but 24% for a WD01 mixture, a factor of 1.5. Only the V band was computed; far-infrared and submillimeter polarized *emission* was not a topic of the paper at all.

1.4 The implementation that survives in the tree

The transport code was MoCafe with forced first scattering and peeling-off. Dichroism was not applied by solving an extinction matrix; polarization was simply incremented at each interaction site. Two consequences follow directly: dichroic extinction never feeds back into the optical depth, and birefringence (the $Q, U \leftrightarrow V$ conversion) is absent entirely.

The actual code is still present at ~/MoCafe/Galaxy_pol/dichroic.f90. Lines 75–89 read:

```
select case(par%dichroic_method)
  case (2)
    pol = par%pol_dichroic * tau**(3.0_wp/4.0_wp)
  case (3)
    pol = par%pol_dichroic * tau**(0.48_wp)
  case (1)
    pol = par%pol_dichroic * tau**(0.48_wp) * sint**2
  case default
    pol = par%pol_dichroic * tau**(3.0_wp/4.0_wp) * sint**2
end select
T1 = cos2p * pol
```

```
T2 = sin2p * pol
photon%Q = photon%Q + T1
photon%U = photon%U + T2
```

with the default amplitude `pol_dichroic = 0.013` set in `Galaxy_pol/define.f90:210`. Two distinct physical defects are visible here.

(i) The τ dependence is a fit, and the wavelength dependence is missing. The correct form for a uniform slab of aligned grains is set by the polarized optical depth $\Delta\tau = n_d C_{\text{pol}} s$, giving

$$p = \tanh(\Delta\tau), \quad (2)$$

which reduces to $p \simeq \Delta\tau$ only in the optically thin limit. Neither $\tau^{0.75}$ nor $\tau^{0.48}$ is that function. More seriously, C_{pol} carries the entire wavelength dependence of dichroism, and it is absent from the expression above: this formulation cannot reproduce a Serkowski curve even in principle, because it has no λ argument.

(ii) The accumulation is additive where the physics is multiplicative. The extinction matrix mixes I into Q , so the correct operation is a matrix exponential acting on the Stokes vector. Incrementing Q and U by a fixed amount at each site agrees with that only while $\tau \ll 1$; the two diverge once $\tau \gtrsim 1$, which is precisely the regime a thick edge-on disk model is built to probe.

2. What has changed since 2018

Observations. NGC 891 now has far-infrared polarimetry. SOFIA/HAWC+ observed it at $154\mu\text{m}$ (Jones et al. 2020, AJ 160, 167; 2020AJ...160...167J), and a deeper analysis was published by Kim et al. (2023, AJ 165, 223; 2023AJ...165...223K). The conclusion of the latter is directly in tension with the 2018 setup: the low polarization fraction observed in the disk, together with its narrow dispersion, requires many turbulent decorrelation cells along each line of sight. A smooth, purely toroidal or purely vertical field cannot produce that combination. Any redo therefore needs a turbulent field component as a first-class ingredient, not an afterthought.

Methodology. Two papers have since addressed exactly the technical obstacles the 2018 implementation stepped around. Baes et al. (2019, A&A 630, A61; 2019A&A...630A...61B) analyzed the optical-depth problem in a dichroic medium, i.e. how to sample path lengths when the extinction coefficient depends on the polarization state being transported. Peest et al. (2023, A&A 673, A112; 2023A&A...673A...112P) provided a full 3D Monte Carlo radiative transfer framework for aligned spheroidal grains, together with analytic benchmarks. Both are directly usable.

3. Polarized optical properties

3.1 The minimum required set

Three cross sections beyond the scalar ones are needed: $C_{\text{ext}}^{\parallel}$ and $C_{\text{ext}}^{\perp}$ for dichroic extinction, $C_{\text{abs}}^{\parallel}$ and $C_{\text{abs}}^{\perp}$ for polarized thermal emission, and C_{circ} for birefringence (the phase retardation that converts linear into circular polarization). In practice these collapse into the combinations C_{ext} , C_{pol} , C_{circ} defined below.

3.2 Modified picket fence approximation

The modified picket fence approximation (MPFA) of Draine & Hensley (2021b, ApJ 919, 65; 2021ApJ...919...65D, Eqs. 11–17) reduces the orientation dependence to two reference orientations. For an oblate spheroid with symmetry axis $\hat{a}$, writing Θ for the angle between the electric field and $\hat{a}$,

$$C_E(\Theta) \simeq \cos^2 \Theta C_E(0) + \sin^2 \Theta C_E(90), \quad (3)$$

and identically for the magnetic-field-referenced C_H . Hensley & Draine (2023, ApJ 948, 55; hereafter HD23, 2023ApJ...948...55H) then write the random-orientation and polarized cross sections as

$$C_{\text{ran}} = \frac{1}{3} [C_E(0) + C_E(90) + C_H(90)], \quad (4)$$

$$C_{\text{pol}}^{\text{oblate}} = \frac{1}{2} [C_H(90) - C_E(90)], \quad (5)$$

their Eqs. 2 and 3 respectively. The accuracy of Eq. (3) is excellent for $2\pi a/\lambda \lesssim 1$ and degrades to roughly 10% near $a \sim \lambda$, but after integration over a realistic size distribution the residual error is a few percent (DH21b).

3.3 Partial alignment and the Rayleigh reduction factor

Partial alignment enters through a single scalar, the Rayleigh reduction factor

$$R \equiv \frac{3}{2} \langle \cos^2 \beta \rangle - \frac{1}{2}, \quad (6)$$

which is the same quantity HD23 call f_{align} . With θ the angle between the magnetic field and the line of sight, the partially aligned cross sections are

$$Q_{\text{abs}}(\theta) = \langle Q \rangle + R(Q_{\parallel} - Q_{\perp}) \left(\frac{1}{3} - \frac{\sin^2 \theta}{2} \right), \quad (7)$$

$$Q_{\text{abs,pol}}(\theta) = -R(Q_{\parallel} - Q_{\perp}) \frac{\sin^2 \theta}{2}. \quad (8)$$

The structural point for implementation is that the entire θ dependence factorizes into the single function $\sin^2 \theta$. Only two wavelength-by-size tables need to be stored, the orientation-averaged one and the polarized one; the geometry factor is applied at run time in the transport loop. No angular dimension is added to the tables.

The corresponding observables are given by HD23 Eq. 6 for polarized extinction and HD23 Eq. 11 for polarized emission, the latter including the integral $\int dT (dP/dT)$ over the stochastic temperature distribution.

3.4 Sign convention

For an oblate spheroid $\hat{a}$ is the *short* axis, and alignment places $\hat{a} \parallel B$. Transmitted starlight is therefore polarized parallel to the plane-of-sky field B_{sky} , while thermal emission is polarized perpendicular to it. Getting this backwards produces a 90° error that is easy to miss in an all-sky map and impossible to miss in a comparison against HAWC+ vectors.

3.5 The asset already in the tree

The single most useful finding of this audit is that no new T-matrix computation is required to get started. The file `data/astrodust/q_DH21Ad_P0.20_Fe0.00_1.400.dat.gz` is not an orientation average. It contains three separate orientation blocks for each of Q_{ext} , Q_{abs} , Q_{sca} , on the native $169 \text{ sizes} \times 1129$ wavelengths grid. Its header states verbatim:

```
3 orientations: jori=1: k || a;; jori=2: k perp a, E || a;
                jori=3: k perp a, E perp a
```

so Eqs. (4) and (5) can be evaluated directly from it.

This has been verified numerically, not merely asserted. Values built from this table reproduce the HD23 released `polarized_extinction.dat` at $0.55 \mu\text{m}$ to four significant figures (ratio 1.0000), and a fit of the alignment function to the same data returns $a_{\text{align}} = 0.07492 \mu\text{m}$, $\alpha = 1.799$, $f_{\text{max}} = 1.0$, identical to HD23 Table 1.

3.6 Wavelength coverage of the polarized optics

Two different objects have two different coverages, and confusing them is easy.

The polarized *extinction* optics — C_{pol} (polarized absorption, which drives the emission), $C_{\text{pol,ext}}$ (dichroism) and $C_{\text{bir,ext}}$ (birefringence) — are tabulated on the full DH21 wavelength axis, all 1129 nodes from 0.0912 to $39810 \mu\text{m}$. Nothing about them is restricted to a band.

The angle-resolved *scattering* matrices of §10. are stored at five wavelengths, 0.36 , 0.44 , 0.55 , 0.64 and $0.79 \mu\text{m}$. That is a deliberate scope, not an incomplete sweep. Polarized radiative transfer is essentially never run on a whole wavelength grid: it is run at the few wavelengths an observation was made at, and five optical bands cover the update this work exists for. What the design has to guarantee is not that every wavelength is precomputed, but that a wavelength one later needs *can* be computed — and it can, by naming it on the generator’s command line (`sedust_polarization_implementation.pdf`, §7 and §8).

3.7 Dichroism below the Lyman limit

The polarized tables stop at $0.0912 \mu\text{m}$ because the DH21 wavelength axis does. A model grid can be extended shortward of it, and when it is, the dichroic and birefringent entries of the extreme-ultraviolet block are currently *zero* — and that zero is a documented deficit, not a physical statement. The distinction matters because a vanishing polarization is a plausible-looking asymptote, so it can be mistaken for a result.

It is not one. First-principles size integrals over the aligned astrodust population give, for the alignment-weighted ratio,

$$\frac{|C_{\text{pol,ext}}|}{C_{\text{ext}}} = \begin{cases} 1.26 \times 10^{-3}, & \lambda = 0.0912 \mu\text{m} \text{ (the seam)}, \\ 3.64 \times 10^{-3}, & 20.6 \text{ eV}, \\ 1.6 \times 10^{-4}, & 100 \text{ eV}, \end{cases} \quad (9)$$

so the dichroism *rises* by a factor 2.9 above the seam before it decays, and it does so with the sign *opposite* to the optical band; the reversal falls near $0.106 \mu\text{m}$. The birefringent ratio $|C_{\text{bir,ext}}|/C_{\text{ext}}$ decreases monotonically from 7.2×10^{-3} at the seam and changes sign near 51 eV.

Zero would be the correct limit only where both $x \gg 1$ and $|m-1| \ll 1$. Neither holds: $|m-1|$ is 0.765 at the seam, 0.564 at 20 eV, and drops below 0.1 only above ~ 50 eV, while the grains that carry the signal sit at $x = 5-50$.

One structural point follows and is worth stating on its own. A sphere has *exactly* zero dichroic extinction and *exactly* zero birefringence, by symmetry. Scalar extreme-ultraviolet optics computed on the volume-equivalent sphere therefore could not have produced these entries at any accuracy: the grain shape has to be retained to obtain them at all, which is why the band needs the orientation-resolved spheroid calculation rather than an extension of the Mie table.

3.8 Index convention traps

The orientation index ordering is not consistent across Draine’s own distributions, and this is a live source of silent sign errors. Draine & Fraisse (2009, ApJ 696, 1; 2009ApJ . . . 696 . . . 1D) use one ordering; the astrodust table uses another. Table 1 gives the mapping.

Table 1: Orientation index conventions. The same three physical orientations are numbered differently in two files from the same source.

Source	$k \perp a, E \parallel a$	$k \perp a, E \perp a$	$k \parallel a$
DF09 (2009ApJ . . . 696 . . . 1D)	$j = 1$	$j = 2$	$j = 3$
q_DH21Ad_*	jori = 2	jori = 3	jori = 1

A third file in the same tree, qlib_gra_D16MGemt_1.400, was for a time read as though it carried a third convention, with jori = 1 taken for the *random-orientation average*. It does not: it uses the same Mishchenko ordering as q_DH21Ad_*, so jori = 1 is $k \parallel a$ there as well, and *none* of the three indices is the random-orientation average. That average has to be built, $\langle Q \rangle_{\text{random}} = \frac{1}{3}Q_{\parallel} + \frac{2}{3}Q_{\perp} = \frac{1}{3}\text{jori}2 + \frac{2}{3}\text{jori}3$, which is what sed/src/q_graphite_d16.f90 now does on load; taking jori = 1 for it instead overstates Q_{abs} by $\sim 22\%$ in the Rayleigh limit, because that orientation carries no $Q_{\parallel}$ at all. A reader routine still has to dispatch on the file identity rather than on the index alone, since DF09 and the astrodust tables genuinely differ.

3.9 HD23 alignment assumptions

For reference, the alignment model adopted by HD23 is: oblate spheroids with $b/a = 1.4$, porosity 0.20, and

$$f_{\text{align}}(a) = \frac{f_{\text{max}}}{1 + (a_{\text{align}}/a)^{\alpha}}, \quad a_{\text{align}} = 0.0749 \mu\text{m}, \quad \alpha = 1.80, \quad f_{\text{max}} = 1.0, \quad (10)$$

with zero polarized contribution from PAHs. The axial ratio $b/a = 1.4$ is not a free aesthetic choice: it is the minimum departure from sphericity that the observed starlight polarization efficiency integral of DH21b will tolerate.

4. What alignment theory hands to radiative transfer

4.1 The interface is one number

Alignment theory, however elaborate, delivers exactly one scalar per grain radius to the transfer solver: $R(a)$. It factorizes into internal alignment (the angle β between the short axis $\hat{a}_1$ and the angular momentum J) and external alignment (the angle ξ between J and B),

$$R = \left\langle \frac{3 \cos^2 \beta - 1}{2} \cdot \frac{3 \cos^2 \xi - 1}{2} \right\rangle, \quad (11)$$

following Hoang et al. (2024, ApJ 965, 183; 2024ApJ...965...183H, Eq. 5). The practical reduced form used in that same paper (their Eq. 12) is a two-state model,

$$R = \begin{cases} 0, & a < a_{\text{align}}, \\ f_{\text{high-J}} Q_X^{\text{high-J}} + (1 - f_{\text{high-J}}) Q_X^{\text{low-J}}, & a \geq a_{\text{align}}, \end{cases} \quad Q_X^{\text{high-J}} = 1, \quad Q_X^{\text{low-J}} \simeq 0.3. \quad (12)$$

4.2 The analytic alignment radius

The reference analytic chain for a_{align} is Hoang, Tram & Lee (2021, ApJ 908, 218; 2021ApJ...908...218H). The RAT efficiency is approximated by

$$\bar{Q}_\Gamma \simeq \begin{cases} 8(\bar{\lambda}/a)^{-3}, & a \lesssim \bar{\lambda}/2.7, \\ 0.4, & a \gtrsim \bar{\lambda}/2.7, \end{cases} \quad (13)$$

the infrared rotational damping coefficient by

$$F_{\text{IR}} \simeq 0.038 U^{2/3} a_{-5}^{-1} n_3^{-1} T_1^{-1/2}, \quad (14)$$

and the suprathermal rotation condition $\omega_{\text{RAT}} = 3\omega_T$ then yields

$$a_{\text{align}} \simeq 0.055 \hat{\rho}^{-1/7} \left(\frac{n_3 T_1}{\gamma_{-1} U} \right)^{2/7} \left(\frac{\bar{\lambda}}{1.2 \mu\text{m}} \right)^{4/7} (1 + F_{\text{IR}})^{2/7} \mu\text{m}. \quad (15)$$

Inside a cloud the local field is reddened and attenuated, which is captured by

$$U = \frac{U_0}{1 + 0.42 A_V^{1.22}}, \quad \bar{\lambda} = \bar{\lambda}_0 (1 + 0.27 A_V^{0.76}), \quad \bar{\lambda}_0 = 1.3 \mu\text{m}. \quad (16)$$

4.3 Rotational disruption

Rotational disruption (RATD) removes large grains whose spin stress exceeds the tensile strength S_{max} . The important structural point is that disruption defines a *band*, not a cutoff: there is an upper bound $a_{\text{disr,max}}$ as well as a_{disr} , because sufficiently large grains again spin too slowly to be disrupted. Implementing RATD as a simple truncation of the size distribution is wrong.

4.4 Magnetically enhanced alignment and its 2025 revision

Magnetically enhanced RAT alignment (MRAT) is parameterized by the magnetic relaxation ratio δ_{mag} . Giang et al. (2023) adopt a step function, $f_{\text{high-J}} = 0.25, 0.5, 1$ for $\delta_m < 1, 1-10$, and > 10 respectively. That last branch is the one to be careful with.

Hoang (2025, ApJ 994, 115) revises the criterion to

$$\delta_{\text{mag,cri}} = (1 + F_{\text{IR}}) \frac{2 - q^{\text{max}}}{q^{\text{max}}}, \quad (17)$$

showing that the familiar “ $\delta_m > 10$ implies perfect alignment” rule was the special case $F_{\text{IR}} \ll 1$. For $U \sim 10^3$ one has $F_{\text{IR}} \sim 60$, so the required δ_{mag} rises by roughly the same factor. Any implementation that keeps the 2016-era step function will therefore overestimate alignment in strong radiation fields, exactly where the polarized emission is brightest. The same paper finds that the attainable maximum of $f_{\text{high-J}}$ is about 0.45 for superparamagnetic grains and about 0.22 for ordinary paramagnetic grains, so the value 1.0 used by POLARIS is optimistic. It also proposes the environment classifier $U/(n_1 T_2)$: values ≤ 1 are collision-dominated, values > 1 are radiation-dominated, in which case the slow-rotation branch has $f_{\text{high-J}}^{\text{slow}} = 0$.

Note the division of labor between these references: the authoritative closed-form expressions for a_{align} and a_{disr} remain those of Hoang et al. (2021). Hoang (2025) does not supply new closed forms; it recasts the problem in a dimensionless parameter space.

4.5 Angle convention

In this literature γ denotes the angle between the *mean magnetic field and the line of sight*, not a plane-of-sky angle. Polarization is maximal at $\gamma = 90^\circ$. Mixing this up with a position angle is a common and quiet error.

4.6 Which axis do grains align with?

Everything above assumes the alignment axis is B . That holds only while Larmor precession is the fastest precession the grain undergoes. Lazarian & Hoang (2019, 2019ApJ . . . 883 . . 122L) set the comparison out explicitly. A grain with a Barnett magnetic moment precesses about B on a timescale (their Eq. 4)

$$t_L = \frac{4\pi}{5} \frac{g e \mu_B}{\hbar} \rho_s s^{-2/3} a_{\text{eff}}^2 B^{-1} \chi(0)^{-1} \approx 1.3 \hat{\rho} \hat{s}^{-2/3} a_{-5}^2 \hat{B}^{-1} \hat{\chi}^{-1} \text{ yr}, \quad (18)$$

with $\hat{B} = B/5 \mu\text{G}$ and $\hat{\chi} = \chi(0)/10^{-4}$. Anisotropic radiation drives a competing precession about k (their Eqs. 12–14),

$$t_{\text{rad,p}} = \frac{2\pi}{\Omega_p}, \quad \Omega_p = \frac{\bar{u}_{\text{rad}} \bar{\lambda} a_{\text{eff}}^2}{I_{\parallel} \omega} \gamma |\bar{Q}_\Gamma|, \quad (19)$$

$$t_{\text{rad,p}} \approx 1.1 \times 10^2 \hat{\rho}^{1/2} \hat{s}^{-1/3} a_{-5}^{1/2} \hat{T}_d^{1/2} \left(\frac{u_{\text{rad}}}{u_{\text{ISRF}}} \right)^{-1} \left(\frac{\bar{\lambda}}{1.2 \mu\text{m}} \right)^{-1} \left(\frac{\gamma |\bar{Q}_\Gamma|}{0.01} \right)^{-1} \text{ yr}. \quad (20)$$

The criterion is simply which is shorter: if $t_{\text{rad,p}} < t_L$, “the radiation direction rather than that of magnetic field becomes the axis of alignment” and the grain is in the k-RAT regime rather than B-RAT.

Two consequences matter for the present purpose. First, the paper’s own estimate is that both the radiative and the mechanical precession rates “are significantly smaller than the rate of Larmor precession,” which “ensures that over

most of the volume of the interstellar medium the alignment is happening in respect to magnetic field.” For a galaxy-scale model such as NGC 891 the B-RAT assumption is therefore safe, and Eq. (18) versus Eq. (20) can be evaluated as a diagnostic rather than carried as a branch in the solver. The regime to watch is the immediate vicinity of bright sources, where $u_{\text{rad}}/u_{\text{ISRF}}$ is large enough to push $t_{\text{rad,p}}$ below t_{L} .

Second, $t_{\text{L}} \propto \chi(0)^{-1}$, so iron inclusions that raise the susceptibility also push the transition further away: the same magnetic enhancement that produces efficient MRAT alignment (§4.4) simultaneously protects the B-RAT assumption. Conversely, detecting a k-RAT signature would constrain $\chi(0)$, which is the observational program that paper proposes. Note also that Eq. (20) uses $|\bar{Q}_{\Gamma}| \approx 2.3(\lambda/a_{\text{eff}})^{-3}$ for $\lambda > 1.8a_{\text{eff}}$ and 0.4 otherwise, whose coefficient and transition point differ from the parameterization quoted in Eq. (13). The two are alternative fits to the same DDA results and agree to within tens of percent; use one consistently rather than mixing them.

4.7 The polarization hole

The mechanism is a chain of three steps, each of which the formulas above already contain: increasing A_V lowers U (Eq. 16), lower U raises a_{align} (Eq. 15), and once a_{align} exceeds the largest grain present, alignment vanishes and polarized emission goes to zero. The complete-loss condition scales as $A_{V,\text{max}} \simeq 24.7(\dots)$, evaluating to roughly 15.9, 6.9, and 3.0 mag for $n_{\text{H}} = 10^4, 10^5, 10^6 \text{ cm}^{-3}$ with $a_{\text{max}} = 0.25 \mu\text{m}$. Grain growth pushes the transition to larger A_V , which is why the observed depth of the hole is a joint constraint on alignment and on growth rather than on either alone.

5. Polarized transfer and its implementation

5.1 The Stokes transfer equation

The transfer of the Stokes vector through an aligned medium is governed by a 4×4 extinction matrix. In the form given by POLARIS I (Reissl, Wolf & Brauer 2016, A&A 593, A87; 2016A&A...593A...87R, their Eq. 7),

$$\frac{d}{ds} \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix} = -n_d \begin{pmatrix} C_{\text{ext}} & C_{\text{pol}} \cos 2\varphi & C_{\text{pol}} \sin 2\varphi & 0 \\ C_{\text{pol}} \cos 2\varphi & C_{\text{ext}} & 0 & C_{\text{circ}} \sin 2\varphi \\ C_{\text{pol}} \sin 2\varphi & 0 & C_{\text{ext}} & -C_{\text{circ}} \cos 2\varphi \\ 0 & -C_{\text{circ}} \sin 2\varphi & C_{\text{circ}} \cos 2\varphi & C_{\text{ext}} \end{pmatrix} \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix} + j, \quad (21)$$

where φ is the position angle of the projected field and j is the polarized emissivity.

Typographical error in the published matrix. The printed (4,4) element in that paper is C_{pol} ; the correct entry is C_{ext} , as written in Eq. (21) above. The reasoning is structural: V must suffer the same total extinction as I , Q , U , and only C_{circ} , appearing antisymmetrically, may rotate U into V . The POLARIS source code implements it correctly, so this is an error in the paper rather than in the code.

5.2 How POLARIS assembles the matrix

The assembly happens in `DustMixture.cpp::calcEmissivityExt`, whose non-zero off-diagonal assignments are:

```

K[0][1] = K[1][0] = Cpol * cos2phi;
K[0][2] = K[2][0] = Cpol * sin2phi;
K[1][3] = Ccirc * sin2phi;
K[2][3] = -Ccirc * cos2phi;
K[3][1] = -Ccirc * sin2phi;
K[3][2] = Ccirc * cos2phi;
    
```

The cross sections themselves absorb the alignment factor in `DustComponent.cpp:calcCrossSections`, as

$$C_{\text{pol}} = \frac{1}{2} R (C_{\text{ext}}^{\perp} - C_{\text{ext}}^{\parallel}) \sin^2 \theta, \quad (22)$$

$$C_{\text{pabs}} = \frac{1}{2} R (C_{\text{abs}}^{\perp} - C_{\text{abs}}^{\parallel}) \sin^2 \theta. \quad (23)$$

A single scalar R thus scales the dichroic and the birefringent amplitudes simultaneously, which is the concrete expression of the statement in §4. that the alignment interface is one number.

5.3 Prefer the closed-form solution to adaptive integration

If the density and the alignment direction are constant within a cell, $\exp(-Kns)$ has a closed form (Peest et al. 2023A&A...673A.112P): I and Q evolve through cosh and sinh of $C_{\text{pol}} ns$, U and V through cos and sin of $C_{\text{circ}} ns$, with an overall factor $e^{-C_{\text{ext}} ns}$. POLARIS instead integrates Eq. (21) with an adaptive Runge–Kutta–Fehlberg 4(5) scheme, applying the error test to each of the four Stokes components with $\epsilon_{\text{err}} = 10^{-8}$. For a cell-based grid the closed form is strictly better: it is exact, it is cheaper, and it removes a tolerance parameter. Peest et al. (2023) also publish analytic benchmark problems, which their MCpol implementation reproduces to within 0.1%; those benchmarks should be adopted as regression tests.

5.4 The three-pass structure and what it buys

POLARIS runs three passes, and the division of labor is the single most useful architectural lesson available:

1. Monte Carlo thermal equilibrium, with alignment ignored.
2. Monte Carlo radiation field, storing a *vector* u_{λ} and the anisotropy measure $\gamma_{\lambda} = |\sum u_i| / \sum |u_i|$.
3. Observables: alignment switched on, Stokes transfer along rays.

During the Monte Carlo passes the alignment is deliberately randomized (`DustMixture.cpp:652--656`). The consequence is worth stating explicitly, because it removes what looks like the largest obstacle: *polarized observables do not require the Monte Carlo transport itself to be polarized*. What is given up is the feedback of dichroic extinction on the radiation field, which is an approximation but a controlled one.

5.5 The radiation field must be vectorial

RAT torque depends on the anisotropy of the local radiation field, so a scalar mean intensity is insufficient: alignment simply cannot be computed from J_{λ} alone. What is required per cell and wavelength is the vector sum as well as the

scalar one,

$$\bar{u}_\lambda = \sum_i |\vec{u}_{\lambda,i}|, \quad \gamma_\lambda = \frac{|\sum_i \vec{u}_{\lambda,i}|}{\sum_i |\vec{u}_{\lambda,i}|} \in [0, 1], \quad (24)$$

with γ_λ the anisotropy degree that enters the torque. This has a hard corollary. The modified random walk (MRW) destroys directional information by construction and therefore cannot be used in the pass that feeds alignment. It remains available in the thermal pass.

5.6 Torque efficiency tables are not needed

A pleasant surprise: POLARIS's `calcAlignedRadii` has the tabulated route (`getQrat`) commented out and uses the power-law approximation of Eq. (13) instead. Equivalent results are therefore obtainable without computing or storing a T-matrix torque-efficiency table, which removes what would otherwise be a substantial Phase-A style computation.

5.7 Stochastic heating and polarization

POLARIS supports the combination, using the Camps et al. (2015) formulation, and the structure is simple: the temperature probability weighting $P(T)$ multiplies I and Q, U identically. Two consequences follow. Within a single grain size the polarization fraction is *independent* of $P(T)$; the temperature distribution affects the net polarization only through the relative weighting of different sizes.

Physically the two populations barely overlap in any case. Alignment sets in near $a_{\text{align}} \simeq 0.075 \mu\text{m}$, whereas stochastic heating dominates for $a \lesssim 0.01\text{--}0.05 \mu\text{m}$. The observational counterpart is the non-detection of polarization in the $3.4 \mu\text{m}$ C–H feature, $0.06\% \pm 0.13\%$ (Chiar et al. 2006, ApJ 651, 268; 2006ApJ...651..268C), consistent with the HD23 assumption of zero polarized PAH emission.

Nevertheless the correct treatment is essentially free here, and that is an argument specific to this code base. SEDust already solves $P(T)$; carrying the polarized emission through the same integral costs one additional accumulator. It is worth noting that the Hoang group papers all use equilibrium temperatures only; Tram et al. (2025, GRADE-POL) states explicitly that the temperature distribution was discarded. Doing it properly would be a genuine, if modest, improvement over the reference implementations.

6. Current state of the codes

6.1 SEDust

Status note (2026 July 19): stage 1 of Table 2 is implemented and verified. The paragraphs below are kept in the past tense where they describe the starting point, and the achieved state is summarized at the end of this subsection. §8. likewise records what was actually done.

SEDust was in better shape than expected. The polarized cross-section table `q_DH21Ad` was already present with all three orientation blocks (§3.), and its correctness has been verified against the HD23 release. The `tmatrix/` library now provides both the random-orientation solve (`tmatrix_eval`, and `tmatrix_eval_scattering_matrix` when the

angular expansion is wanted) and Mishchenko's fixed-orientation amplitude routine AMPL, so SEDust can also regenerate the orientation-resolved cross sections from the dielectric function itself; the default polarized optics still read the release table, but they no longer have to. A T-matrix run is therefore useful not only for the scattering matrix (§9., now done for five optical bands) and for exploring other axial ratios or compositions, but for the polarized cross sections directly. The release table carries only Q_{ext} , Q_{abs} and Q_{sca} , so from it C_{circ} is zero; the regenerated table adds a birefringence block (the real part of the forward-amplitude difference) from which C_{circ} follows, certified by Kramers-Kronig against the dichroism.

The structural entry point was `grain_pop_t` in `sed/src/dust_model_mod.f90`: adding a `Cpol(NLAM,NA)` component there propagates naturally through the existing population loop. On the emission side, one polarized accumulator alongside the existing one in `sed_astrodust.f90` completed the change.

What SEDust now provides. The orientation-resolved reader is `sed/src/q_table_jori.f90`. `sed_init` builds two polarized cross sections on the size-distribution grid, `Cpol` (polarized absorption, which drives the polarized emission) and `Cpol_ext` (polarized extinction, which drives the dichroic term of the extinction matrix), together with $f_{\text{align}}(a)$ from the HD23 fit. Three exports follow:

- `dust_emission` takes an optional `lamI_pol` of the same shape and units as `lamI_total`, returning the *intrinsic* polarized emission;
- `../data/astrodust/kext_astrodust_MW.dat` carries the polarized extinction per H as an eighth column, appended so that readers of the first seven columns are unaffected;
- `calc_pol_ext.x` regenerates the comparison against the released `polarized_extinction.dat`.

The reproduction is a median $|\text{ours}/\text{ref} - 1|$ of 0.0296% over the 992 points with $\lambda \geq 0.11 \mu\text{m}$, and the intrinsic polarization fraction of the emission is 19.15% at $850 \mu\text{m}$ and 17.15% at $154 \mu\text{m}$. The calling interface is specified in the SEDust user manual; `docs/mocafe_polarization_handoff.md` records the decisions taken while building this and the limits of the released data.

6.2 MoCafe v2.00

MoCafe already carries the polarization state. `photon_type` has I,Q,U,V (`define.f90:117`), there is a `use_stokes` switch (line 213), and `scatter_dust_stokes` interpolates S_{11} , S_{12} , S_{33} , S_{34} . SEDust is embedded through `dustemis_mod`, and the code passes the Camps et al. (2015) SHG benchmark. The task is already on record: the roadmap contains [TODO, L] Polarized thermal emission from aligned grains. Long-term.

Four things are missing:

1. No magnetic field exists anywhere (a source-wide `grep` returns zero hits). This is the largest single gap.
2. Extinction is scalar: $\tau = \rho_{\text{hokap}} * \text{length}$.
3. `j_tally_mod` accumulates a *scalar* J_{λ} only, and must be vectorized before any self-consistent alignment calculation is possible.

4. There is no polarized emission source term.

6.3 Reference sources in the tree

Two POLARIS checkouts are available for reference: Grain/POLARIS/ (branch master-basic, commit 03280f5) and Grain/NewPOLARIS/ (branch main, commit e508e89; tag v1 = a4fcb4d corresponds to Zenodo 10.5281/zenodo.14199014, the published version, with main ahead only in the README).

NewPOLARIS is the MRAT and RATD extension of Giang, Hoang & Kim (2023). It is *not* merged upstream and was forked from an older layout (Dust.cpp/Dust.h), so a direct diff against the baseline is not possible; only the physics can be ported. What it adds: the ALIG_MRAT, ALIG_kRAT, ALIG_INELASTIC flags; the keywords <number_cluster>, <volume_filling_cluster>, <iron_fraction>, <change_f_highJ>, <inelascity>, <disrupt> RATD; and grid fields holding critical radii for each cell (GRIDadISR, GRIDamaxJB_Lar, GRIDadg_lower/upper), which moves the expensive physics into a preprocessing step. The stock POLARIS has no MRAT, and its ALIG_KRAT is a dead flag: it is connected to neither the parser nor any physics path.

7. Staged proposal

Table 2: Proposed stages, with the status as of 2026 July 19. Stage 2 is the point at which quantitative comparison against SOFIA/HAWC+ becomes possible; stage 1 no longer blocks it.

Stage	Scope	Effort	Status	Content
1	SEDust cross sections	days, low risk	done	Cpol, Cpol_ext and f_{align} built in sed_init; polarized emission accumulator in sed_grain_loop; optional lamI_pol on dust_emission; polarized extinction exported as column 8 of kext_astrodust_MW.dat. Released polarized_extinction.dat reproduced to a median of 0.0296%; intrinsic polarization fraction 19.15% at $850\mu\text{m}$.
2	MoCafe ray tracing	moderate	next	Polarized emission in ray tracing and peel-off, using the Peest closed form. Needs no scattering matrix: the far-infrared albedo is near zero.
3	MoCafe forward MC	moderate	open	Dichroic extinction; keep scalar τ for path sampling and correct by weight. Column 8 supplies C_{pol} .
4	Self-consistent alignment	large	open	Vectorial radiation-field pass; MRW disabled; memory $\times 4$.

Stage 1 (done). C_{pol} , $C_{\text{pol_ext}}$ and f_{align} are built in SEDust, with a single polarized emission accumulator alongside the existing one. Nothing in the solver core changed, and the three astro dust SEDs plus the DL07 SED regress byte-identically. The first validation step, reproducing the released `polarized_extinction.dat`, passes at a median of 0.0296%. The remaining checks, p_{submm} against Planck, $R_{S/V}$, and the Serkowski parameters λ_{max} and K , are now possible from column 8 of `kext_astrodust_MW.dat` and remain to be done.

Stage 2. Polarized emission in the MoCafe ray tracing and peel-off paths, integrated with the closed-form solution of §5.. This is the milestone worth aiming at first: it already allows a quantitative comparison with the HAWC+ $154\mu\text{m}$ maps, which did not exist in 2018.

Stage 3. Dichroic extinction in the forward Monte Carlo. The recommended approach keeps the scalar τ for path-length sampling and applies a weight correction, which sidesteps the circularity analyzed by Baes et al. (2019). At this stage the empirical `dichroic.f90` formula is replaced by Eq. (2) with a wavelength-dependent C_{pol} .

Stage 4. Alignment computed internally. This requires the vectorial radiation-field pass, forbids MRW in that pass, and raises the radiation-field memory by a factor of four. Until then the sensible choice is the SKIRT approach: treat $f_{\text{align}}(a)$ as an input function, with the HD23 fit of Eq. (10) as the default.

8. What the update actually touches

The stages above are stated in physical terms. This section translates them into the specific data structures and routines that would change, based on the state of the two trees as of 2026 July. Line numbers are given where they pin down a unique insertion point; they will drift.

8.1 SEDust, stage 1 (implemented)

This subsection was written as a plan and is kept for the reasoning it records. What was actually built follows each paragraph in italics. The resulting calling interface is specified in the SEDust user manual.

Read the polarized cross sections. The existing loader `load_q_table` (`sed/src/q_table.f90`) reads the flat table produced by `run_tmatrix`, whose columns are `lambda` `a_eff` `Qext` `Qabs` `Qsca` `albedo` `g` `flag`. The HD21 file has a different shape: three orientation blocks, each 1129×169 , written as $((Q(jw, jr, jori), jw=0, 1128), jr=0, 168), jori=1, 3$ for each of Q_{ext} , Q_{abs} , Q_{sca} in turn. Two options exist: add a second reader for the HD21 layout, or extend the existing table format with a `jori` column and regenerate. The first is faster to get running; the second keeps one reader for both the shipped table and any future T-matrix computation, and is the better end state. Whichever is chosen, form

$$Q_{\text{pol}}^{\text{abs}}(\lambda, a) = \frac{1}{2} \left[Q_{\text{abs}}^{(jori=3)} - Q_{\text{abs}}^{(jori=2)} \right], \quad Q_{\text{pol}}^{\text{ext}} \text{ likewise from } Q_{\text{ext}}, \quad (25)$$

and watch the index convention of Table 1.

Done as the first option: `sed/src/q_table_jori.f90` is a second reader for the HD21 layout, leaving `q_table.f90` untouched. It decompresses the shipped `.gz` through `gzip -dc` into a scratch copy, and exposes `qpol_ext`, `qpol_abs`, the random-orientation combinations `qran_*`, the grids `lam_j`, `aeff_j`, and `falign_hd23(a)`.

Carry it on the population. `grain_pop_t` (`sed/src/dust_model_mod.f90`) currently holds `aeff`, `dn`, `Cabs`, `Csca`, `kappB`, `H` and `kappCMB`. Add

```
real(wp), allocatable :: Cpol(:, :) ! (NLAM, NA) polarized absorption
real(wp), allocatable :: falign(:) ! (NA) alignment efficiency
```

and free them in `free_pop` alongside the rest. Filling `falign` from Eq. (10) needs only `aeff`, which the population already carries after the earlier fix that added it.

Done, with `set_pop` taking `Cpol` and `falign` as optional arguments so that populations without polarized optics simply leave them unallocated. The module-level arrays behind them are `Cpol` and `Cpol_ext`; the short name stays on the absorption one because the emission path is written around it.

Accumulate the polarized emission. The emission loop in `sed_grain_loop` (`sed/src/sed_astrodust.f90`) has four accumulation sites, at roughly lines 757, 762, 926 and 938, of the form

```
Jout_local = Jout_local + dn_pop(ir) * P(ii) * Cabs_pop(:, ir) * spec
```

Each gains a companion line that differs only in the cross section and the alignment weight:

```
Jpol_local = Jpol_local + dn_pop(ir) * P(ii) * Cpol_pop(:, ir) &
               * falign_pop(ir) * spec
```

This is the point made in §5.7: because the temperature weight multiplies both accumulators identically, the exact $P(T)$ treatment comes for free, and there is no reason to approximate the polarized emission with an equilibrium temperature. The two sites at 757 and 938 are the equilibrium branches and need the same companion.

Done at every accumulation site, guarded by a single `do_pol` flag that is true only when the caller asked for the polarized output and the population carries both arrays.

Widen the public interface. `dust_emission` (`sed/src/sed_astrodust.f90:2143`) currently returns `lamI_total` and optionally `lamI_chan`. Add an optional `lamI_pol` of the same shape as `lamI_total`. Making it optional keeps every existing caller valid, which matters because MoCafe links this as a library; the same discipline was used for the status argument. Re-export through `dust_lib`.

Done. The signature is now `dust_emission(m, J_lam, lamI_total [, lamI_chan] [, status] [, lamI_pol])`. The polarized extinction took a different route: rather than a second library accessor, it is written as column 8 of `../data/astrodust/kext_astrodust_MW.dat`, because that file is already how a host consumes extinction and `dust_lib` exposes no extinction surface for an accessor to sit beside.

Note that what leaves SEDust is the *intrinsic* polarized emissivity, that is, the quantity before the geometric factor $\sin^2 \gamma$ and before any turbulent reduction. Those depend on the local field direction and belong to the RT side. Keeping the split here is what allows a single pair of tables to serve all viewing geometries.

Verify before going further. Reproduce `data/release/polarized_extinction.dat` by combining Eq. (5) with the alignment fit of Eq. (10) and integrating over the size distribution. This exercises the orientation indexing, the sign convention, the f_{align} fit and the size integration in one shot. It has already been shown to work at $0.55 \mu\text{m}$; extending the check across the full wavelength grid is the natural first commit. Only then are the p_{submm} , $R_{S/V}$ and Serkowski comparisons of §11. meaningful.

Done by `calc_pol_ext.x`, which reproduces the release to a median of 0.0296% and a maximum of 0.94% over the 992 points with $\lambda \geq 0.11 \mu\text{m}$ (the reference itself turns negative below that). Column 8 of the extinction table agrees with that program to the seven digits the ASCII format carries, so the two routes to the same quantity are consistent. The p_{submm} , $R_{S/V}$ and Serkowski comparisons are still open.

8.2 MoCafe, stage 2

A magnetic field has to exist. No field of any kind appears in the current source. A minimal implementation adds a field direction for each cell, either from an analytic model or from an MHD snapshot, alongside the existing density. For NGC 891 the analytic route is the starting point, but the observational result of Kim et al. (2023) means a turbulent component is not optional: a smooth field cannot produce the low polarization fraction and narrow dispersion they report. In practice this means either resolving the turbulence on the grid or carrying a reduction factor F_{turb} per cell, as GRADE-POL does.

Emissivity becomes a Stokes vector. `compute_dustemis(src/dustemis_mod.f90)` currently obtains a scalar spectrum from `dust_emission` or `dust_emission_interp` and stores one emissivity array per cell. With `lamI_pol` available it stores, for each cell,

$$j_I = (\text{as now}), \quad \begin{pmatrix} j_Q \\ j_U \end{pmatrix} = (\lambda I_\lambda)_{\text{pol}} \sin^2 \gamma F_{\text{turb}} \begin{pmatrix} \cos 2\varphi \\ \sin 2\varphi \end{pmatrix}, \quad (26)$$

where γ is the angle between the local field and the line of sight to the observer and φ is the position angle of the field projected on the sky. Both follow from the field direction and the observer direction, so no new physics enters here.

There is a structural consequence worth noting: γ and φ depend on the viewing direction, so j_Q and j_U are not properties of the cell alone in the way j_I is. For a single observer this is immaterial. For the all-sky output of `allsky_mod` it means the angles must be evaluated per direction rather than cached per cell.

Integrate the Stokes vector along the ray. The ray-tracing and peel-off paths (`src/peelingoff_mod.f90`, `src/allsky_mod.f90`) currently accumulate a scalar optical depth through `raytrace_tau` and apply $e^{-\tau}$. Replace this, for the polarized output only, by the segment-by-segment application of the closed-form solution of §5.: within a segment of constant density and field direction the Stokes vector transforms by cosh and sinh of $C_{\text{pol}}ns$ in (I, Q) and by cos and sin of $C_{\text{circ}}ns$ in (U, V) , all scaled by $e^{-C_{\text{ext}}ns}$, with a rotation into and out of the local field frame at each segment boundary. The existing segment walk already provides the geometry; what changes is what is done with each segment. At this point the model produces polarized emission maps that can be compared with the HAWC+ $154 \mu\text{m}$ data directly. Nothing in the forward Monte Carlo has been touched.

8.3 MoCafe, stage 3

The forward Monte Carlo is where dichroism becomes awkward, because the optical depth that governs path sampling is no longer a scalar. The recommendation of §7. is to keep sampling path lengths from the scalar C_{ext} and to apply the polarization as a weight correction on the surviving packet, which preserves the sampling distribution and avoids the circularity that Baes et al. (2019) analyze. The physical cost is that dichroic extinction does not feed back on the radiation field; whether that matters can be measured by comparing stage-3 output with stage-2 output, which is a reason to build them in this order.

This is also the point at which `Galaxy_pol/dichroic.f90` would be retired. Its replacement is Eq. (2) with C_{pol} from Eq. (25), which restores the wavelength dependence that the empirical form cannot represent.

8.4 MoCafe, stage 4

Only here does the radiation-field tally change. `jtally_mod` stores `jt_sum(nlambda,ncell)`, a scalar mean intensity. Radiative torque alignment needs the direction of the radiation as well, so the tally becomes four numbers per cell and wavelength: the scalar sum and the three components of $\sum_i \vec{u}_{\lambda,i}$, from which the anisotropy degree γ_{λ} of Eq. (24) follows. The memory note already in the roadmap, that `jt_sum` is held privately per rank, becomes four times more pressing.

Two further constraints come with it. The modified random walk cannot be used in this pass, since it destroys the directional information. And the alignment solve itself is a preprocessing step over cells, not an inner-loop cost: as NewPOLARIS shows, the sensible structure is to compute a_{align} , a_{disr} and the magnetic radii once per cell, store them, and have the cross-section evaluation do nothing but compare grain size against stored radii.

8.5 Order of work and what each stage buys

Stage 1 is self-contained inside SEDust and verifiable against a published file. Stage 2 produces the first result that could not have been obtained in 2018, because the HAWC+ maps did not exist. Stage 3 removes the empirical formula that is the weakest point of the 2018 model. Stage 4 replaces the assumed $f_{\text{align}}(a)$ with computed alignment, and is the only stage that changes the Monte Carlo itself.

A useful property of this ordering is that each stage is independently publishable and independently falsifiable: stage 1 against `polarized_extinction.dat`, stage 2 against HAWC+, stage 3 against stage 2, and stage 4 against the fixed- f_{align} result of stage 2.

9. What optical polarized RT additionally requires

Everything up to this point is written with the far infrared in view, where the albedo is near zero and a scattering matrix can be skipped (§7., stage 2). Optical starlight polarization is the case the 2018 model actually addressed, and it needs strictly more. This section collects the additional requirements in one place, because they are currently scattered: the fixed-orientation amplitude code is mentioned in §6., C_{circ} in §3. and §5., and the MPFA in §3..

The immediate goal is to update Seon (2018), which worked at a single wavelength (V band, §1.). Concretely, the target is to replace the empirical $P_V = 2.7 \tau_V^{0.48}$ of Eq. (1) by dichroism computed from cross sections. A plausible extension is one or two further optical bands, which is a different thing from a full wavelength grid: the far infrared is not in the present plan at all, and neither is the ultraviolet. The requirements below are ordered on that basis, and items that only bind once the ultraviolet or the far infrared enters are marked as such. Several requirements that look formidable when stated for a full wavelength grid stay modest across two or three optical bands, and the ordering exploits that.

Two consequences of the wider optical scope are worth stating at the outset. A scattering matrix would be needed at each band rather than at one wavelength, which multiplies the computation but not the method: a handful of bands is still far from a grid. And once more than one band is available the wavelength dependence of the polarization becomes an observable in its own right, which is exactly what the empirical formula of Eq. (1) cannot reproduce, so the case for replacing it grows stronger rather than weaker.

9.1 The stage table presupposes data that does not exist yet

Table 2 assigns stages 3 and 4 to MoCafe, which is correct as far as transport is concerned but incomplete as a plan. Those stages consume optical quantities that SEDust did not originally produce, and in one case that no released file contains. Stage 1 delivered C_{pol} and $C_{\text{pol_ext}}$, sufficient for polarized emission and for dichroic extinction; the random-orientation scattering matrix has since been added for five optical bands. Table 3 lists the supply side item by item, with the priority each carries for the single-wavelength update.

Scattering matrix, random orientation (available). This item is done. The T-matrix driver in the tree was originally derived from Mishchenko’s `tmd.lp.f` by wrapping the implicit main program as a subroutine, and in the process the final MATR call was removed and the angular expansion coefficients were dropped from the returned quantities. The MATR routine itself remained in the file, and the call was later restored. That `f77` driver has since been rewritten as the modern-Fortran library in `tmatrix/src/`: the entry point that returns the expansion is `tmatrix_eval_scattering_matrix`, which fills a `tmatrix_scattermat_t` (`al1...al4`, `be1`, `be2`, `lmax`, `nmax_tm`), and MATR itself is `SCATMAT_FROM_MOMENTS` in module `scattering_matrix_expansion`. Mishchenko’s unmodified sources are kept under `tmatrix/reference/upstream/` so that every migrated routine stays checkable line by line against them; nothing there is compiled.

`tmatrix/driver/run_scattermat.f90` uses it to accumulate the single-size matrices over the HD23 astro dust size distribution with $C_{\text{sca}} dn/n_H$ weighting and to expand the result onto a scattering-angle grid. The output is `tmatrix/output/scattermat_astrodust_P0.20_Fe0.00_1.400.dat`: one block per wavelength, 181 rows at $\theta = 0, 1, \dots, 180^\circ$, with the six independent elements a randomly oriented particle with a plane of symmetry possesses, in the column order $F_{11}, F_{22}, F_{33}, F_{44}, F_{12}, F_{34}$. The normalization is Mishchenko’s, $\frac{1}{2} \int_{-1}^{+1} F_{11} d \cos \theta = 1$, so absolute units follow by multiplying by the C_{sca} quoted in each block header. This already improves on the 2018 model, whose phase matrix was spherical Mie.

Five wavelengths are stored, 0.36, 0.44, 0.55, 0.64 and $0.79 \mu\text{m}$, approximately $UBVRI$. That covers the scope of the update as planned — the V band first, with one or two more optical bands possible, and no far-infrared or ultraviolet use foreseen. The full 1129-point sweep costs about 1.7 hours and would produce mostly wavelengths outside that scope,

Table 3: Ingredients that far-infrared polarized transfer can do without and optical polarized transfer cannot. “Supplier” is the code that must produce the quantity; “Priority” is judged against the single-wavelength (V band) goal, not against a full wavelength grid.

Ingredient	Supplier	Priority	Status, and why optical needs it
$Z(\theta)$ at V , random orientation	SEDust	done	Computed by <code>run_scattermat.x</code> and stored for five optical bands, $UBVRI$. Scattering dominates the optical opacity, so the phase matrix sets the scattered polarization; in the far infrared the albedo is near zero and Z never enters.
f_{align} functional form	either	high	HD23 fit assumed. Optical polarization is carried by grains near $a \sim a_{\text{align}}$, so the shape of the cutoff is a first-order effect, and it is what sets how the polarization varies from band to band; far-infrared polarization comes from grains well above it, where all forms agree.
$Z(\lambda, \theta)$, aligned orientation	SEDust	done	Now built: the full fixed-orientation Mueller matrix $Z(\theta_i; \theta_s, \phi)$ is formed at all scattering angles from AMPL and size-integrated over the aligned population, in five optical bands (§10.). Aligned spheroids scatter anisotropically with respect to B , which spherical Mie cannot represent.
C_{circ}	SEDust	done (optics)	Birefringence converts U into V . Absent from the release, but now computed as the real part of the forward-amplitude difference and stored in the regenerated table; Kramers-Kronig certified. Returning it through <code>dust_extinction</code> and consuming it in transfer is still to do.
Averaging convention	SEDust	low	Mixed, exact average for C_{abs} vs. MPFA for C_{pol} . The disagreement is 0.080% median across the optical, so it does not bind on a programme confined to optical bands. It becomes an obstacle in the ultraviolet, where the median is 0.21% and the maximum 9.53%. Deliberately deferred in v1.20; see the paragraph below.

so it was not run. It requires no code change should the scope widen: `./run_scattermat.x all` computes the whole grid, or an explicit list of wavelengths a subset, at about 5.5 s per wavelength.

What remains missing is the *aligned* scattering matrix, and C_{circ} ; both are treated in the two paragraphs below. The distinction matters: the random-orientation matrix answers how an unaligned grain population scatters, which is what a

first optical model needs for the scattering source term, but it carries no information about the orientation of the grains with respect to B .

The cutoff shape is the physics specific to the optical (needed). Eq. (10) is a fitted function, and §7. treats it as an input rather than a computed quantity. In the far infrared that is harmless, because the emission is dominated by grains several times larger than a_{align} , where any reasonable form has saturated at f_{max} . At V the polarization is produced by grains of size comparable to the wavelength, which is the same size range as a_{align} itself, so the result responds directly to whether the cutoff is a step or a smooth roll-off. The observational expression of this is that Serkowski’s λ_{max} is set largely by the alignment size threshold rather than by the size distribution. Truong et al. (2025ApJ...981...83T) make the same point from the other side, noting that the step and smooth forms are distinguishable only in the optical. This item does not become easier at a single wavelength and keeps its priority.

Scattering matrix, aligned orientation (built). The full orientation-dependent amplitude matrix is evaluated by AMPL, in `tmatrix/src/fixed_orientation_amplitude.f90`. Retaining it at all scattering angles and assembling the sixteen Mueller elements — the step that was missing — is now done, and the result is size-integrated over the aligned population into the phase matrix $Z_{\text{al}}(\theta_i; \theta_s, \phi)$ and the extinction matrix $K(\theta_i)$. The cost turned out to be storage rather than algorithm, as anticipated: the shipped file spans five optical bands at 138 MB (§10.). There is no shortcut through published data — the HD21 release carries integrated Q_{ext} , Q_{abs} and Q_{sca} only, with no scattering matrix at any orientation — so this had to be computed from the T-matrix directly.

Birefringence came free with the aligned extinction, as predicted. C_{circ} is absent from the release but needs no separate computation. In the forward-scattering amplitude the difference $S_{\theta\theta} - S_{\phi\phi}$ carries dichroism in its imaginary part and birefringence in its real part, so the fixed-orientation amplitude run already done for C_{pol} yields C_{circ} from the same output. Keeping the real part is now implemented: the regenerated table carries it as a 4th block, and it is certified by Kramers-Kronig against the dichroism to a median $\sim 0.1\%$ over the interior of the grid, the only check available since the release publishes no birefringence. `dust_extinction` now returns it through an optional `Cbir_ext` argument, and the aligned scattering table carries the same birefringence in its $K(\theta_i)$ (§10.); what remains is using it in the $U \leftrightarrow V$ transfer term, which is the host’s task.

The averaging conventions do not match, but not yet critically. SEDust builds C_{abs} from the exact orientation average of its own T-matrix route and C_{pol} from the MPFA combination of Eq. (5). The two conventions disagree by a median of 0.022% in the far infrared and 0.080% in the optical, and by a median of 0.21% with a maximum of 9.53% in the ultraviolet. Across the optical the 0.080% figure is far below every other uncertainty in the problem, so this obstructs neither the single-band update nor an extension to two or three optical bands. It becomes an obstacle as soon as polarized extinction is quoted in the ultraviolet, where the polarization curve is measured. Note that resolving it changes C_{abs} , and therefore breaks the byte-identical regression of the existing SEDs; that is an engineering consequence to plan around, not a reason to leave two conventions in one model permanently.

The regenerated orientation-resolved table (the first-principles route added after v1.20; see `sedust_polarization_implementation.pdf`) is the concrete way to unify them. It supplies C_{pol} from the ex-

act fixed-orientation T-matrix, $\frac{1}{2}(Q_3 - Q_2)$, rather than from the MPFA approximation of Eq. (5), so both channels rest on the same exact optics and the mixed convention is removed outright, at the price of the FIR emission dichroism disagreeing with the release by the same few parts in 10^4 . It is opt-in for exactly this reason, to keep the regression baseline until that step is taken deliberately.

For v1.20 the decision was taken explicitly to defer it. The scope of the update lies wholly inside the range where the disagreement is 0.080% or less, and unifying the convention would spend the regression baseline against which everything else in the release was validated, in exchange for accuracy nothing in the current plan needs. The deferral is a statement about ordering, not about correctness: two orientation averages in one model is still wrong, and the first ultraviolet polarized-extinction result is the point at which it must be fixed rather than postponed again.

9.2 Recommended order

For the update as scoped, one band first and possibly two or three optical bands after, the realistic sequence is:

1. Compute the V-band dichroic cross section from physics and retire the empirical formula, using the Cpol_ext that stage 1 already produces.
2. Test the sensitivity to the f_{align} functional form, step versus smooth.
3. Bring the random-orientation scattering matrix into the transfer. The supply side of this step is complete: run_scattermat.x already provides $Z(\theta)$ in five optical bands, and widening that set costs another run rather than another method. What remains is consuming it.
4. Only then bring in the aligned scattering matrix (its supply is now built, §10.; what remains is consuming it), C_{circ} , the averaging convention, and any extension beyond the optical.

The point worth emphasizing is that step 1 on its own is already the substantive advance over Seon (2018): it replaces a fitted power law that has no wavelength argument and no alignment physics with a cross section computed from the grain model, which is exactly the defect identified in §1.. Step 1 needs no new T-matrix run, because Cpol_ext exists and has been validated against the HD23 release. Step 3 is placed after step 2 because it is the point at which the comparison against the release is partly lost, the release containing no scattering matrix; entering locally computed territory is better done once the cheaper questions have been answered.

9.3 What the Hoang group assumes

Since the alignment inputs are assumptions rather than derived quantities, it is useful to record the values in current use by the group that produces most of this literature. Table 4 collects them. Most of these carry over to a single-wavelength calculation unchanged: $f_{\text{high-J}}$, Q_X , Q_I , the iron-inclusion parameters and F_{turb} are all wavelength-independent alignment parameters, so they enter the V-band Rayleigh reduction factor exactly as they enter a submillimeter one. Only the axial ratio and the f_{align} shape have consequences that differ from band to band.

Four features of this table are worth drawing out, because each is a place where our own model can differ deliberately rather than by accident.

Table 4: Parameter values adopted in the Hoang-group modeling papers. Where a paper scans a parameter, the scanned range is given.

Quantity	Value	Source and remark
$Q_X^{\text{high-J}}, Q_X^{\text{low-J}}$	1, $\simeq 0.3$	2024ApJ...965...183H Eq. 12; standard case, suprathermal rotation taken to give perfect internal alignment
same, slow internal relaxation	0.15, 0.05	2025A&A...703A.192T; wrong internal alignment enters as -0.1
Q_J (external alignment)	1 in both states	2024ApJ...965...183H; assumed perfect, so the whole reduction of R comes from internal alignment. Stated by the authors as optimistic
$f_{\text{high-J}}$	0.25, 0.5, 1.0	2023MNRAS.520.3788G Eq. 31, a step function in δ_{mag} ; used as model labels for pure RAT, intermediate and full MRAT. A footnote there gives 0.25–0.7 for compact irregular grains, making 1.0 an upper scenario
$f_{\text{high-J}}^{\text{fast}}$, attainable maximum	~ 0.45 (SPM), ~ 0.22 (PM)	2025ApJ...994...115H; the value 1.0 used by POLARIS is therefore optimistic (§4.4)
N_{cl} , iron atoms in a cluster	$\sim 5\text{--}10^3$	2025ApJ...980...105G; reproducing ALMA data needs $10^2\text{--}10^4$ together with $a \geq 10\mu\text{m}$ and inelastic relaxation
ϕ_{sp} , cluster volume filling	0.1	2025ApJ...980...105G, with the iron fraction f_p
Axial ratio	1.4 default; 1.6, 2.0, 3.0 scanned	2023ApJ...948...55H for the default; reproducing Pipe-109 needs 1.4 at low A_V and 2–3 above $A_V = 14$ (2025A&A...703A.192T)
F_{turb}	0.98–0.91	$F_{\text{turb}} = \frac{1}{2}[3\langle\cos^2\Delta\theta\rangle - 1]$; Pipe-109 values, 2025A&A...703A.192T
u_{MMP}	$8.64 \times 10^{-13} \text{ erg cm}^{-3}$	normalization of $U = u_{\text{rad}}/u_{\text{MMP}}$
T_d	$16.4 \text{ K} (a/0.1 \mu\text{m})^{-1/15} U^{1/6}$	equilibrium temperature only
f_{align} shape	step, or $f_{\text{max}}[1 - e^{-(0.5a/a_{\text{align}})^3}]$	step in POLARIS; the smooth form is 2020ApJ...896...44L, used by GRADEPOL on its semi-analytic path

Negative Q_X is a physical statement, not a bookkeeping sign. In the conservative slow-relaxation model the wrong internal alignment state carries $Q_X = -0.1$, and a negative Rayleigh reduction factor means the polarization direction flips by 90° . This is the origin of the observed pattern in which P is parallel to B at millimeter wavelengths and

perpendicular to it in the submillimeter. An implementation that clamps R to non-negative values will silently delete this effect.

External alignment is assumed, not computed. With $Q_J = 1$ in both the low- J and high- J states, the entire departure of R from unity in these papers comes from internal alignment. The papers say so; it is an acknowledged optimistic approximation rather than a hidden one. It also means that quoted values of R are upper bounds.

The radiation field normalization is used loosely. The group writes $U = u_{\text{rad}}/u_{\text{MMP}}$ with $u_{\text{MMP}} = 8.64 \times 10^{-13}$ erg cm^{-3} throughout, without distinguishing the 1.0×10^{-13} and 1.65×10^{-13} dilution factors of the Mathis field and without mentioning the CMB. A survey of eleven papers from the group found zero that make either distinction. For SEDust this is a place where the existing radiation-field handling is already more careful, and the difference should be stated rather than absorbed.

The temperature treatment is equilibrium throughout. No paper in that survey of eleven cites an astro dust heat capacity, and at least one (2026ApJ...1005...195T) uses astro dust cross sections together with a silicate-based temperature scaling. Tram et al. (2025) state explicitly that the temperature distribution was discarded. As noted in §5.7, SEDust already solves $P(T)$, so the correct treatment costs one accumulator here and is a genuine improvement over the reference implementations.

9.4 A working prescription

Given those assumptions, the practical question is what to compute and in what order. The following is written against the current state of our own code rather than against POLARIS.

1. Start with a fixed f_{align} . Adopt the HD23 fit of Eq. (10) with $a_{\text{align}} = 0.0749 \mu\text{m}$, $\alpha = 1.80$, $f_{\text{max}} = 1$, and do not compute R at all. This is already implemented and verified (§6.). The first optical step is then not to compute alignment but to swap in the smooth form of Table 4 and measure how much the polarization curve moves. If it moves little, the assumed form is not the limiting uncertainty; if it moves a lot, that settles the priority.

2. Replace the empirical dichroism at V . Use Cpol_ext in Eq. (2) in place of Eq. (1), with the alignment weight applied as in Eq. (22). This is the deliverable: the polarization is then set by the grain model and the field geometry rather than by a fitted exponent, and the accumulation becomes multiplicative as §1. requires. No new optical data is needed.

3. Only then compute alignment. Computing R rather than assuming it needs the analytic chain of §4., Eqs. (15) and (16), evaluated on the local radiation field, which in turn requires the vectorial tally of §5.5. $f_{\text{high-}J}$ follows from δ_{mag} , either through the step function of Giang et al. (2023) or the revised criterion of Eq. (17). This is stage 4 of Table 2 and should not be attempted before steps 1 and 2 have shown what the fixed- f_{align} model cannot do.

4. Climb the validation ladder in order. Reproduce the released polarized extinction (done, §8.); then the Serkowski $\lambda_{\max}$ and K ; then $p_V/E(B-V)$ against the $\geq 13\%$ /mag requirement of §11., not the classical 9%/mag, which a model that under-polarizes will pass; then $R_{S/V}$; then the HAWC+ and Planck comparisons. The ordering matters because each rung uses only quantities the previous rungs have already validated.

9.5 A constraint that applies only to a multi-wavelength extension

One result deserves to be recorded here even though it is out of scope for the single-wavelength update, because it will be the first difficulty encountered as soon as the scope widens. The Pipe-109 analysis found that no single $a_{\max}$ fits all bands: 0.19–0.25 μm in R , 0.25–0.35 μm in H , and 1–3 μm at 850 μm , and that the axial ratio has to grow with A_V , from 1.4 at low extinction to 2–3 above $A_V = 14$ (2025A&A...703A.192T). Fitting optical, near-infrared and submillimeter polarization at once therefore strains the size distribution and the grain shape before it strains the alignment physics. Updating Seon (2018) at V alone does not run into this: a single band constrains one combination of size distribution and shape, and no tension can arise between bands that are not being fitted. The constraint is listed so that it is not rediscovered as a surprise, and so that it is not mistaken for a present obstacle.

An honest limit. Following this prescription does not close the gap in polarization amplitude. Our intrinsic polarization fraction tops out at 19.15% at 850 μm against the Planck maximum of 22% (Table 6). That shortfall is a property of the $b/a = 1.4$ oblate spheroid and will not be removed by better alignment physics or a better solver. It is the same pressure that leads the Hoang group to raise the axial ratio to 2–3 when fitting real sight lines, and it belongs with the open shape question of §12..

10. The aligned scattering matrix, as built

*Status note (2026 July 20). The item this section describes — $Z(\lambda, \theta)$ for aligned orientation, listed as “later” in earlier drafts and now marked **done** in Table 3 — is built and verified. Its implementation is documented in `sedust_polarization_implementation.pdf` §8 and its calling interface in the SEDust user manual; this section records the physics and the place it takes in the transfer.*

The random-orientation matrix of §9. answers how an *unaligned* population scatters. An aligned grain scatters differently depending on how the incident ray is inclined to its symmetry axis, and that dependence is the scattering-side counterpart of the dichroic extinction the model already supplies. SEDust now computes it for the astrodust spheroid, from the same fixed-orientation T-matrix amplitude used for the oriented cross sections.

10.1 Why the phase matrix carries three angles

For a randomly oriented particle with a plane of symmetry the scattering matrix depends on the single scattering angle Θ and has six independent elements (the $F_{11}, F_{22}, F_{33}, F_{44}, F_{12}, F_{34}$ of §9.). Fix the grain — place its symmetry axis $\hat{a}$ along z — and that reduction no longer holds: the matrix depends on the incidence direction relative to the axis *and* on the full scattering direction. Writing θ_i for the polar angle of incidence from the axis and (θ_s, ϕ) for the scattering

direction, axial symmetry leaves only the azimuth *difference* ϕ , so

$$Z = Z(\theta_i; \theta_s, \phi), \quad 16 \text{ elements, } \mu\text{m}^2 \text{ sr}^{-1}. \quad (27)$$

Sixteen elements rather than six is the price of dropping the orientation average, and the extra argument θ_i is exactly the inclination of the ray to B that carries the aligned-scattering information.

10.2 The meridional basis

Stokes vectors are referred to Mishchenko's meridional basis of each propagation direction *in the grain frame*, whose z axis is the alignment axis (the local B): $\hat{e}_1 = \hat{\theta}$ (call it v), $\hat{e}_2 = \hat{\phi}$ (h), with $Q = I_v - I_h$. This is the basis into which the fixed-orientation amplitude matrix S is naturally rotated, so no adaptor stands between the analytic dipole limit and the T-matrix regime. At $\theta_i = 90^\circ$ the v polarization is $E \parallel \hat{a}$ (the jori = 2 orientation of §3.) and h is $E \perp \hat{a}$ (jori = 3), which ties the extinction matrix at edge-on incidence back to the jori cross sections and fixes the sign of $C_{\text{pol}} = \frac{1}{2}[C_H(90) - C_E(90)]$ of Eq. (5). The transfer code rotates the photon Stokes vector into this grain frame at each scattering event and back out afterward.

10.3 Exact symmetries

Three symmetries of the oblate spheroid reduce the stored angular ranges and, more usefully, are exact numerical checks on the matrix construction:

- **Azimuth mirror.** The plane through the axis and the incidence direction is a symmetry plane, so $\phi \rightarrow 360^\circ - \phi$ leaves Z unchanged except that the two off-diagonal 2×2 blocks (elements 13, 14, 23, 24, 31, 32, 41, 42) change sign. Only $\phi \in [0, 180^\circ]$ need be stored.
- **Equatorial mapping.** Under $z \rightarrow -z$, $\theta_i \rightarrow 180^\circ - \theta_i$ maps onto the stored range with $\theta_s \rightarrow 180^\circ - \theta_s$, ϕ unchanged, and the same off-diagonal-block sign flip (the diagonal blocks keep their sign). Only $\theta_i \in [0, 90^\circ]$ need be stored; physically this encodes that the alignment axis is headless.
- **Axisymmetry at $\theta_i = 0$.** When the ray travels along the axis the problem is axisymmetric: Z becomes independent of ϕ up to a frame rotation and takes the six-element block-diagonal form. This is a check, not a saving.

All three were verified numerically (to $\sim 10^{-15}$ for the equatorial mapping, a few parts in 10^7 for the azimuth mirror) before being used to reconstruct the unstored octants.

10.4 Aligned plus unaligned, exactly

Partial alignment is handled by the same decomposition used for the extinction: a perfectly aligned fraction $f_{\text{align}}(a)$ plus a randomly oriented remainder,

$$Z_{\text{total}}(\theta_i; \theta_s, \phi) = \int da n(a) [f_{\text{align}}(a) Z_{\text{oriented}}(a) + (1 - f_{\text{align}}(a)) F_{\text{random}}(a)]. \quad (28)$$

SEDust stores the aligned integral $Z_{\text{al}} = \int da n(a) f_{\text{align}}(a) Z_{\text{oriented}}(a)$, the random-orientation matrix F_{tot} of every grain, and its aligned-weighted twin $F_{\text{ref}} = \int da n(a) f_{\text{align}}(a) F_{\text{random}}(a)$. Because Eq. (28) is *linear* in f_{align} , a cell alignment

written as $f_{\text{align}}^{\text{cell}} = \eta f_{\text{align}}^{\text{ref}}$ with one scalar η scales the aligned part exactly, $Z_{\text{al}}^{\text{cell}} = \eta Z_{\text{al}}$, and the unaligned remainder is $F_{\text{unal}} = F_{\text{tot}} - \eta F_{\text{ref}}$ (in absolute units, after restoring each matrix's C_{sca}). Three stored integrals therefore give the scattering optics of any η with no re-integration and no approximation — the Rayleigh reduction factor R of §4. enters as this η . A genuinely different profile shape requires regenerating the integrals over size.

10.5 The extinction matrix from forward amplitudes

The 4×4 extinction matrix of Eq. (21) needs $C_{\text{ext}}(\theta_i)$, the dichroism $C_{\text{pol}}(\theta_i)$, and the birefringence $C_{\text{circ}}(\theta_i)$ at the ray's inclination to the field. These are read from the *forward* amplitude at each θ_i — the same amplitude call the scattering matrix uses — with C_{ext} from its imaginary part (the optical theorem) and C_{circ} from its real part, exactly as §3. anticipated for the free birefringence. Stored on the θ_i grid, the extinction matrix is thus *exact at every propagation angle*.

This supersedes the interpolation the three-orientation table forces. The jori table gives C_E only at $\psi = 0$ and 90° , so a transfer code has to interpolate between them with the modified picket fence $\sin^2 \psi$ law of Eq. (3); away from small size parameters that law is only accurate to about ten percent. The forward-amplitude route removes the interpolation. In the Rayleigh limit it *reduces* to the $\sin^2$ law exactly: the electric-dipole cross sections obey

$$C_v(\theta_i) = \sin^2 \theta_i C(E \parallel \hat{a}) + \cos^2 \theta_i C(E \perp \hat{a}), \quad C_h(\theta_i) = C(E \perp \hat{a}), \quad (29)$$

so $C_{\text{pol}}(\theta_i) = \frac{1}{2}[C_h - C_v] = -\frac{1}{2} \sin^2 \theta_i [C(E \parallel \hat{a}) - C(E \perp \hat{a})]$, which is Eq. (8) with $R \rightarrow \eta$. The dipole $K(\theta_i)$ reproduces Eq. (29) to machine precision, and the T-matrix $K(\theta_i)$ follows it to a few parts in 10^4 at small x , which is the anchor that the exact route and the interpolated one agree where they must.

10.6 Where each quantity enters the Peest transfer

The Monte Carlo formalism of Peest et al. (2017, 2017A&A...601A...92P), extended to aligned spheroids, dichroism and birefringence by Peest et al. (2023, 2023A&A...673A.112P), and the scattering-plus-dichroism transfer of Seon (2018, 2018ApJ...862...87S) are the destination. Each SEDust quantity maps onto one step of it (Table 5); the division of labor is that SEDust returns the matrices in the grain frame and MoCafe performs the frame rotations, the direction sampling, the peel-off, and the $\exp(-K\tau)$ transfer.

The far-infrared and submillimeter milestone needs none of the scattering rows: there the albedo is essentially zero and only the $K(\theta_i)$ row and the emission vector are used. The scattering rows are what the optical update of Seon (2018) additionally requires, and they are now supplied.

11. Validation targets

Correction to the classical limit. The last row deserves emphasis, because using the traditional value will let a bad model pass. The classical upper bound $p_V/E(B-V) \leq 9 \text{ \%/mag}$ is obsolete; a model that under-polarizes will satisfy it comfortably. The post-Planck requirement is $\geq 13 \text{ \%/mag}$, and the HD23 target value is 0.13/mag.

Table 5: Which SEDust quantity enters which step of the Peest et al. Monte Carlo transfer. All are read from the aligned scattering table through the library API; $\theta_i = \arccos(\hat{k} \cdot \hat{B})$ and the cell alignment scale η are the only inputs.

Transfer step	SEDust quantity	Role
Optical depth / attenuation	$K(\theta_i) = \eta K_{\text{al}}(\theta_i) + (C_{\text{ext}}^{\text{iso}} - \eta C_{\text{ext,ref}})I$	The 4×4 extinction operator of Eq. (21): C_{ext} on the diagonal, C_{pol} in the IQ block (dichroism), C_{circ} anti-symmetric in the UV block (birefringence). The aligned piece is direction-resolved; the isotropic remainder avoids double-counting the aligned population.
Albedo, population split	<code>scattering_cross_sections</code> $\rightarrow C_{\text{sca,al}}(\theta_i), C_{\text{sca,unal}}$	The single-scattering albedo and the probability that an interaction is with the aligned versus the unaligned population.
Direction sampling	$Z_{\text{al},11}(\theta_i; \theta_s, \phi)$ or $F_{\text{unal},11}(\Theta)$	The sky brightness the scattered direction is drawn from — the aligned phase function in the grain frame, or the random-orientation one in the scattering plane.
Stokes update at a scatter	full Z_{al} (aligned) or the six-element F_{unal} (unaligned)	Transform the Stokes vector, with meridional rotations into and out of the grain frame for the aligned case; the unaligned case works in the scattering plane as the existing scatmat file already does.
Peel-off to the observer	$Z_{\text{al}} / F_{\text{unal}}$ toward $\hat{n}_{\text{obs}}$	Evaluate the same phase matrix at the forced scattering geometry and weight by the albedo.

Table 6: Acceptance targets for a polarized dust model.

Quantity	Value	Source
$\lambda_{\max}$	$0.55 \mu\text{m}$	2009ApJ...696...1D
Serkowski K	$-0.01 + 1.66 \lambda_{\max}$	1992ApJ...386...562W
$p_{\max}$ at 353 GHz	$22.0^{+3.5}_{-1.4} \%$	2020A&A...641A...12P
$R_{S/V}$	4.2 ± 0.2	2015A&A...576A.106P
$R_{P/p}$	$5.4 \pm 0.2 \text{ MJy/sr}$	2015A&A...576A.106P
Polarization spectrum, $250 \mu\text{m}$ – 3 mm	flat to $\lesssim 10\%$	HD23
$p_V/E(B-V)$	$\geq 13 \text{ \%/mag}$	2020A&A...641A...12P, 2019A&A...624L...8P

Values already computed here. Three numbers were obtained during this audit and are available as immediate checks:

- $Q_{\text{pol}}/Q_{\text{abs}} = 0.2790$ in the Rayleigh limit for $b/a = 1.4$, independent of size and wavelength. This is a pure geometry check and should be reproduced exactly.
- $p_{\text{submm}} = 19.5\%$ with f_{align} applied, against the Planck value of 22.0% .
- $R_{S/V} = 3.85$ under a single-temperature approximation, against 4.2 observed and 4.31 as the HD23 target.

The third of these is the most informative, and it points to a natural first task. The 11% shortfall in $R_{S/V}$ is controlled by the relative emission weighting across grain sizes, and that weighting is exactly what SEDust’s $T(a)$ and $P(T)$ solution provides. Recomputing $R_{S/V}$ with the full temperature treatment rather than a single T is therefore both a physics improvement and a sharp test of the new code path.

12. Unverified items

The following are stated as open, not as established. They are listed so that none of them is silently promoted to fact in later work.

1. **F_{IR} coefficient normalization.** Baseline POLARIS uses $1.40\text{e}10$ where NewPOLARIS uses $6.14\text{e}9$. Whether this is a difference of unit system or a genuine disagreement was not determined. Since F_{IR} enters both Eq. (15) and Eq. (17), the factor matters and must be resolved before either is trusted.
2. **Origin of the $R_{S/V}$ gap.** Whether 3.85 versus 4.31 is entirely attributable to the single-temperature approximation, or partly to a difference in the definition of the ratio, is not established.
3. **Grain shape is an open choice, not a settled one.** Guillet et al. (2018) favor prolate grains with $a/b = 3$ plus porosity, while HD23 use a single-composition oblate spheroid with $b/a = 1.4$. These are mutually incompatible solutions to overlapping data, and adopting either is a modeling decision that should be recorded as such.

13. References

- Baes, M., et al. 2019, A&A, 630, A61. 2019A&A...630A...61B
- Bianchi, S., Ferrara, A., & Giovanardi, C. 1996, ApJ, 465, 127. 1996ApJ...465...127B
- Camps, P., et al. 2015, A&A, 580, A87. 2015A&A...580A...87C
- Chiar, J. E., et al. 2006, ApJ, 651, 268. 2006ApJ...651...268C
- Draine, B. T. 2003, ApJ, 598, 1017. 2003ApJ...598.1017D
- Draine, B. T., & Fraisse, A. A. 2009, ApJ, 696, 1 (DF09). 2009ApJ...696...1D
- Draine, B. T., & Hensley, B. S. 2021a, ApJ, 909, 94 (DH21a). 2021ApJ...909...94D
- Draine, B. T., & Hensley, B. S. 2021b, ApJ, 919, 65 (DH21b). 2021ApJ...919...65D
- Draine, B. T., & Lee, H. M. 1984, ApJ, 285, 89. 1984ApJ...285...89D
- Draine, B. T., & Li, A. 2007, ApJ, 657, 810 (DL07). 2007ApJ...657...810D
- Giang, N. C., et al. 2025, ApJ, 980, 105. 2025ApJ...980...105G
- Giang, N. C., Hoang, T., & Kim, J.-G. 2023, MNRAS, 520, 3788 (2023MNRAS.520.3788G). NewPOLARIS MRAT and RATD extension, Zenodo 10.5281/zenodo.14199014
- Guillet, V., et al. 2018, A&A, 610, A16. 2018A&A...610A...16G
- Hensley, B. S., & Draine, B. T. 2023, ApJ, 948, 55 (HD23). 2023ApJ...948...55H
- Hoang, T. 2025, ApJ, 994, 115 (2025ApJ...994...115H).
- Hoang, T., et al. 2024, ApJ, 965, 183. 2024ApJ...965...183H
- Hoang, T., Tram, L. N., & Lee, H. 2021, ApJ, 908, 218. 2021ApJ...908...218H
- Jones, T. J., et al. 2020, AJ, 160, 167. 2020AJ...160...167J
- Kim, J., et al. 2023, AJ, 165, 223. 2023AJ...165...223K
- Lazarian, A., & Hoang, T. 2019, ApJ, 883, 122. 2019ApJ...883...122L
- Lee, H., et al. 2020, ApJ, 896, 44. 2020ApJ...896...44L
- Peest, C., Camps, P., Stalevski, M., Baes, M., & Siebenmorgen, R. 2017, A&A, 601, A92. 2017A&A...601A...92P
- Peest, C., et al. 2023, A&A, 673, A112. 2023A&A...673A.112P
- Planck Collaboration Int. XXI 2015, A&A, 576, A106. 2015A&A...576A.106P
- Planck Collaboration 2019, A&A, 624, L8. 2019A&A...624L...8P
- Planck Collaboration 2020, A&A, 641, A12. 2020A&A...641A...12P

- Reissl, S., Wolf, S., & Brauer, R. 2016, A&A, 593, A87 (POLARIS I). 2016A&A...593A...87R
- Seon, K.-I. 2018, ApJ, 862, 87. 2018ApJ...862...87S; arXiv:1806.06101
- Serkowski, K., Mathewson, D. S., & Ford, V. L. 1975, ApJ, 196, 261. 1975ApJ...196..261S
- Tram, L. N., Hoang, T., & Lazarian, A. 2025, A&A, 703, A192 (2025A&A...703A.192T). GRADE-POL I.
- Truong, B., et al. 2025, ApJ, 981, 83. 2025ApJ...981...83T
- Truong, B., et al. 2026, ApJ, 1005, 195. 2026ApJ...1005..195T
- Whittet, D. C. B., et al. 1992, ApJ, 386, 562. 1992ApJ...386..562W
- Whittet, D. C. B., et al. 2008, ApJ, 674, 304. 2008ApJ...674..304W
- Wood, K. 1997, ApJ, 477, L25. 1997ApJ...477L..25W
- Wood, K., & Jones, T. J. 1997, AJ, 114, 1405. 1997AJ...114.1405W
- Zubko, V., Dwek, E., & Arendt, R. G. 2004, ApJS, 152, 211. 2004ApJS...152..211Z